



Unifying Global and Local Features

A Modern Approach to Sports Video Analysis

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- **Applications:**
 - ▶ Highlight generation
 - ▶ Video analytics and broadcasting automation
 - ▶ Downstream temporal retrieval

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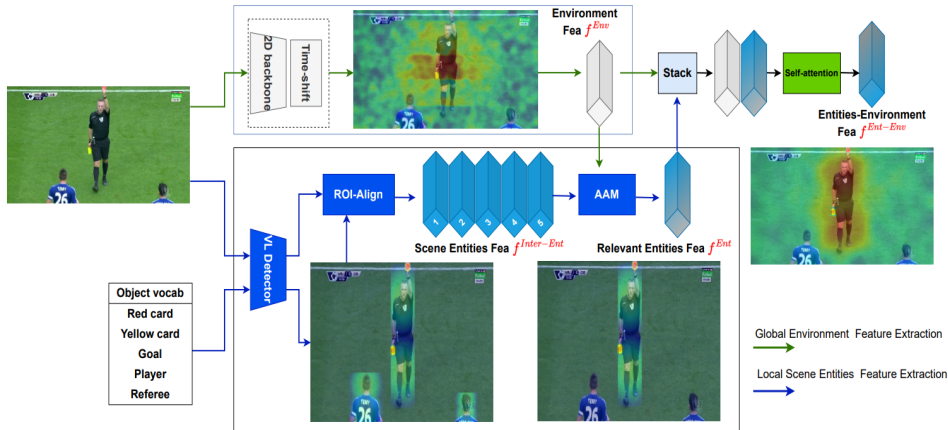
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- End-to-end methods (e.g., E2E-Spot): lighter but still global-only
- **Need**: explicit modeling of both global and local scene elements

UGL Architecture

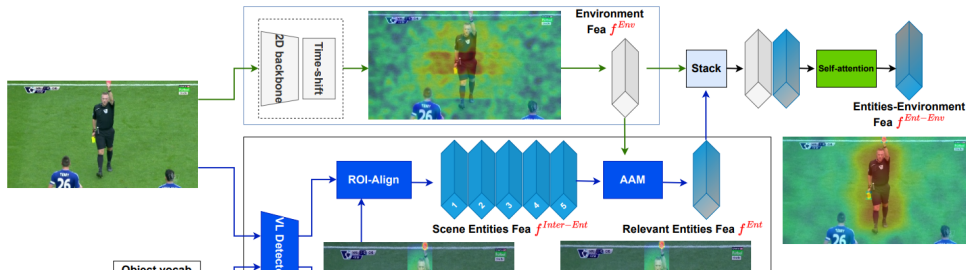
Global Environment Feature



UGL Architecture

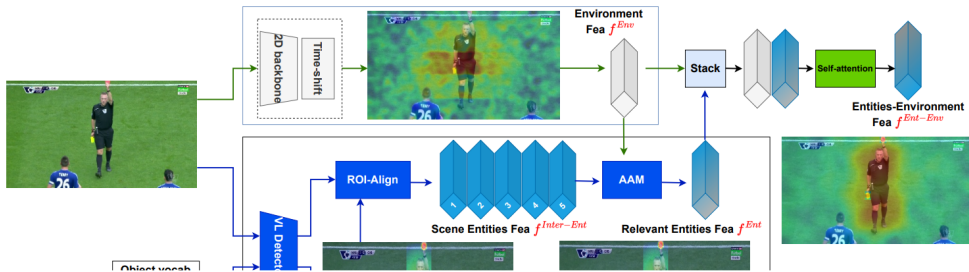
Global Environment Feature

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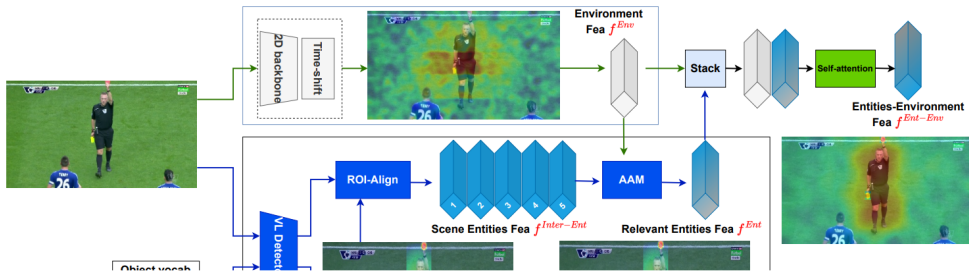
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- **Backbone:** RegNet-Y (efficient 2D CNN design)
- **Temporal Integration:** Gate-Shift Module (GSM)
 - ▶ Shifts feature maps forward and backward along temporal axis
 - ▶ Shares information across frames
 - ▶ Lightweight alternative to 3D CNNs



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 - ▶ Lightweight alternative to 3D CNNs
- **Output:** Global environment feature vector per frame capturing spatial and short-term temporal context



Why GLIP?

Global features alone often ignore small but crucial entities (cards, ball, etc.), especially under class imbalance

- **GLIP**: Grounded Language-Image Pre-training (training-free detector)
- **Input**: Frame image + text prompt listing sports entities
 - ▶ *“ball, yellow card, red card, referee, ...”*
- **Mechanism**:
 - ▶ Language encoder processes text (BERT)
 - ▶ Visual encoder produces image proposals (Swin backbone)
 - ▶ Cross-modal multi-head attention aligns proposals with textual entities

Adaptive Attention Mechanism (AAM)

Problem

Not all detected entities are equally relevant for the current action

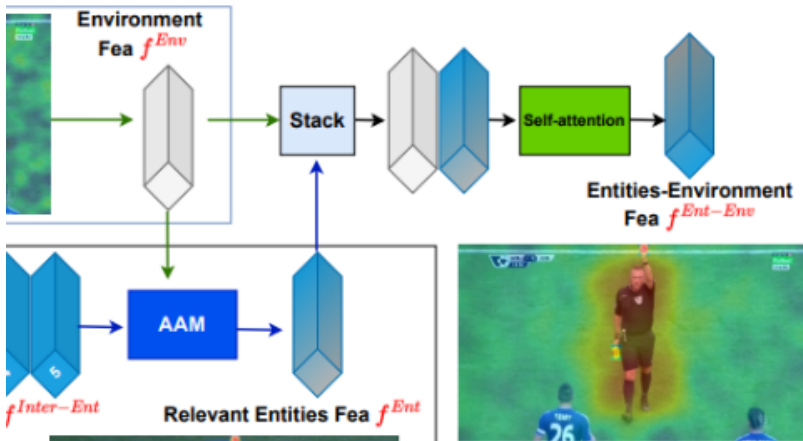
AAM Solution

Takes both local entity features and global environment feature:

1. **Hard attention**: Select most promising entities
2. **Soft self-attention**: Fuse them into single compact representation

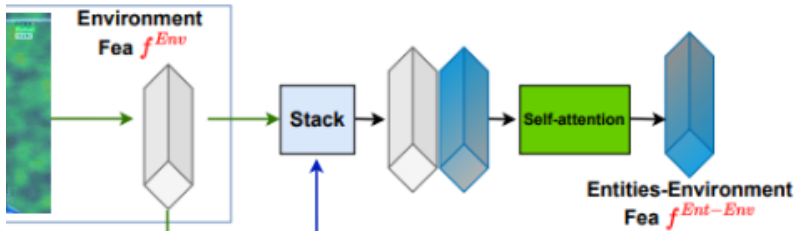
Output

Local relevant scene entities feature emphasizing objects crucial for the action



1. Concatenate:

- ▶ Global environment feature
- ▶ Local relevant entities feature



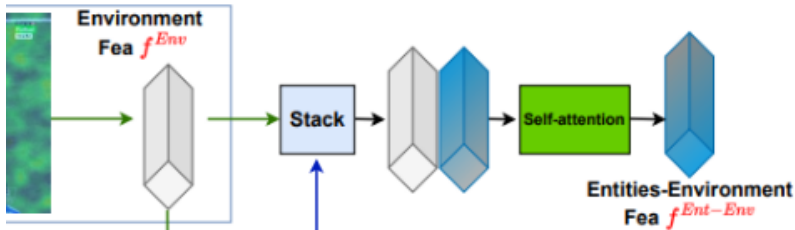
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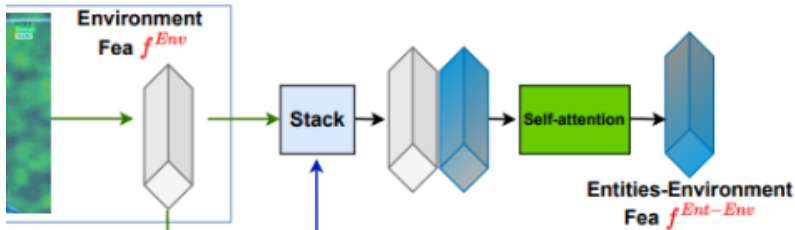
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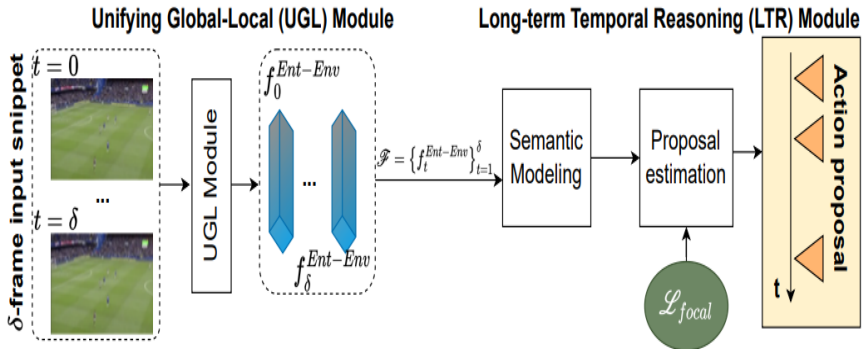
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3. Output: Unified entities and environment representation per frame capturing both background context and key object semantics



Long-Term Temporal Reasoning (LTR)



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Temporal Modeling

- Input: Sequence of fused frame features for T -frame snippet
- Model: 1-layer bidirectional GRU capturing semantic relationships
- Prediction: Per-frame class scores via fully connected + softmax

Inference with Sliding Window

- Slide window over entire video with overlap
- Merge frame-wise scores from GRU for each snippet
- Apply Non-Maximum Suppression (NMS) in time
- Output: Temporally precise event spots

Three Sports Domains

SoccerNet-v2

- ~550 matches
- ~764 hours of video
- >300k labeled spots
- Strong clutter, camera changes

FineDiving

- 3,000 diving clips
- Phase transitions (entry, twist, etc.)

FineGym

- 5,374 gym performances
- 32 spotting classes
- Converted from action segments to timestamps

The Problem

In SoccerNet-v2: background frames dominate; non-background classes $\approx 2\%$ of frames

Solution: Focal Loss

- Down-weights easy examples with high predicted probability
- Focuses learning on hard, misclassified, or minority-class examples
- Includes balancing parameter for class weighting
- Applied per frame over predicted multi-class outputs

mAP

Mean Average Precision per class at fixed temporal tolerance

Average-mAP (A-mAP)

Area under mAP vs tolerance curve across range of tolerances

Tight Average-mAP (T-mAP)

Focuses on tight tolerances from 1 to 5 seconds, better reflects precise spotting capability

Tabelle: SoccerNet-v2 challenge quantitative comparison (T-mAP).

Method	T-mAP (%)	Gap to UGL (%)	Notes
UGL (ours)	69.38	0.00	Top-1 on challenge leaderboard
E2E-Spot	66.73	-2.65	End-to-end baseline
Baidu (two-phase)	49.56	-19.82	Computed from reported margin
Soares (two-phase)	67.81	-1.57	Computed from reported margin

Tabelle: FineDiving and FineGym improvements of UGL over E2E-Spot (no optical flow).

Dataset	Δ mAP of UGL (%)	Comment
FineDiving	+2.4	At selected temporal tolerances
FineGym	+1.3	Consistent gain over baseline

- UGL unifies global environment and local entity features to produce precise, interpretable per-frame representations
- Top result on SoccerNet-v2: **69.38% T-mAP** (Top-1); +2.65% vs E2E-Spot
- Consistent gains on FineDiving (+2.4% mAP) and FineGym (+1.3% mAP)
- Adaptive Attention selects relevant entities; GRU + Focal Loss handles temporal reasoning and class imbalance effectively

